

Predictive Modeling for Course Demand and Revenue Forecasting on EduPro

Research Paper

1. Introduction

EduPro, an online learning platform, currently plans course launches, pricing, and instructor onboarding based on historical intuition rather than quantitative forecasts. This creates business risk: without visibility into which courses are likely to succeed, EduPro cannot confidently decide where to invest in new content or how to price it.

This project introduces predictive intelligence to EduPro's planning process. The objective is to build machine learning models that forecast three outcomes for a course: how many students will enroll, how much revenue it will generate, and how much revenue its category will produce in aggregate. The goal is to move EduPro from reactive reporting to proactive, data-driven planning.

2. Dataset Overview

The dataset consists of four related tables:

- Courses (60 records) - CourseID, category, type, level, price, duration, rating
- Teachers (60 records) - TeacherID, expertise, years of experience, rating
- Transactions (10,000 records) - links users, courses, and teachers, spanning January to December 2025
- Users (3,000 records) - demographic information, not used in core modeling per project scope

The dataset contained no missing values across any table. Courses are evenly distributed across 12 categories (5 courses each), which gives balanced representation but also means each category is a fairly small sample.

3. Exploratory Data Analysis - Key Insights

Several findings shaped the modeling approach:

- 38 of 60 courses (63%) are priced at \$0 (free). This explains why 6,403 of 10,000 transactions (64%) recorded an Amount of \$0 - these are legitimate free enrollments, not missing or corrupted data.
- Free courses generated 6,403 enrollments but \$0 in revenue; paid courses generated 3,597 enrollments but \$911,323 in revenue. Enrollment volume and revenue are therefore driven by different dynamics and should not be treated as proxies for one another.
- For paid courses, Course Revenue is close to CoursePrice multiplied by EnrollmentCount, meaning revenue is largely a mechanical downstream consequence of price and enrollment rather than an independent pattern to discover. Enrollment Count is therefore the more business-critical and analytically interesting target.
- Instructor assignment is not fixed per course: each course was taught by between 7 and 30 distinct teachers across its transactions. Teacher-related features were therefore aggregated as course-level averages (average teacher rating, average teacher experience) rather than treated as a single fixed instructor attribute.

4. Methodology

4.1 Data Preparation

Transactions were aggregated by CourseID to compute EnrollmentCount (transaction count) and CourseRevenue (sum of Amount). Teacher quality metrics were joined to transactions via TeacherID, then averaged per course. These aggregates were merged with the Courses table to produce a single course-level table of 60 rows.

4.2 Feature Engineering

The following engineered features were added to capture non-linear effects that raw numeric values might miss:

- Price bands (Free / Low / Medium / High)
- Duration buckets (Short / Medium / Long / Very Long)
- Rating tiers (Low / Mid / High)
- Instructor experience buckets (Junior / Mid / Senior)

4.3 Preprocessing

Categorical variables (course category, type, level, and the engineered bands/tiers/buckets) were one-hot encoded, expanding the feature set to 29 columns. Data was split 80/20 into training and test sets (random_state=42 for reproducibility).

4.4 Models Trained

Four regression models were trained for each target variable: Linear Regression and Ridge Regression as baselines, and Random Forest Regressor and Gradient Boosting Regressor as advanced, non-linear models capable of capturing feature interactions.

5. Model Results

5.1 Enrollment Count Prediction

Model	MAE	RMSE	R ²
Linear Regression	11.23	14.01	-0.077
Ridge Regression	8.52	10.80	0.360
Random Forest	10.25	12.40	0.157
Gradient Boosting	11.61	13.60	-0.015

Ridge Regression performed best for enrollment prediction ($R^2 = 0.360$, MAE = 8.52 students). This result is honest rather than impressive: with only 60 courses total and 48 in the training set, there is limited data for the model to learn from, and enrollment count is influenced by real-world factors not captured in this dataset (e.g. marketing spend, seasonality, platform placement). Ridge's regularization likely helped it generalize better than the unregularized Linear Regression and the more complex tree-based models, which are more prone to overfitting on small datasets. This is a meaningful and defensible finding: it demonstrates that model selection and honest evaluation matter more than chasing a high R^2 score, and it points directly to a recommendation (see Section 7) that EduPro should collect more courses and additional behavioral features to improve future forecasts.

5.2 Course Revenue Prediction

Model	MAE (\$)	RMSE (\$)	R ²
Linear Regression	3,341.43	4,043.43	0.979
Ridge Regression	3,131.80	3,837.83	0.981
Random Forest	3,463.79	5,489.83	0.961
Gradient Boosting	2,797.35	4,054.81	0.979

All four models explain over 96% of the variance in course revenue, with Gradient Boosting achieving the lowest MAE (\$2,797) and Ridge the highest R^2 (0.981). As noted in Section 3, this strong performance is expected given that revenue for paid courses is closely tied to price and enrollment, both of which are available features.

6. Feature Importance Analysis

Using the Random Forest model trained on Enrollment Count, the following features were most influential:

Rank	Feature	Relative Importance
1	Course Rating	24.0%
2	Course Duration	16.8%
3	Category: Data Science	13.8%
4	Average Teacher Experience	9.6%
5	Average Teacher Rating	7.6%
6	Level: Intermediate	4.1%
7	Price Band: Medium	3.4%
8	Category: Digital Marketing	3.1%

Course quality signals (rating, duration) and category dominate over price band as demand drivers, suggesting that for this dataset, perceived course quality and subject area matter more to enrollment than price positioning alone. Instructor quality (experience and rating combined) accounts for roughly 17% of predictive weight, indicating instructors are a meaningful but secondary factor.

7. Business Recommendations

- Prioritize course quality signals (production value driving higher ratings, appropriately scoped duration) over aggressive discounting, since rating and duration outweighed price band in driving enrollment.
- Data Science stood out as a category-specific driver of enrollment; combined with its strong revenue performance (\$77,052 total, 5 courses), it is a reasonable candidate for continued or expanded investment.
- Free courses drive substantial enrollment volume (6,403 enrollments) but zero direct revenue. A hybrid or freemium pricing model - free introductory content with paid advanced modules - could convert this existing enrollment interest into revenue without abandoning the acquisition benefit of free courses.
- Instructor experience and rating together account for a meaningful share of predictive power; investing in instructor quality (vetting, training, or ratings-based incentives) is a lever EduPro can act on directly.
- Given the modest R^2 for enrollment prediction, EduPro should treat these forecasts as directional guidance rather than precise targets, and should prioritize collecting more courses and additional behavioral data (e.g. marketing spend, traffic source, seasonality) to improve model reliability over time.

8. Conclusion

This project converted EduPro's fragmented historical data - courses, teachers, and transactions - into a unified, forward-looking analytical asset. Four models were trained and evaluated across three business-relevant targets, and the resulting insights were translated into concrete recommendations on pricing strategy, category investment, and instructor quality. While enrollment prediction accuracy is currently limited by dataset size, revenue prediction is strong, and the modeling pipeline, dashboard, and feature importance analysis together give EduPro a repeatable framework it can re-run and refine as more data becomes available - directly addressing the original problem of relying on intuition rather than evidence for course planning decisions.